Output Reference

SWAT+ writes text files with results for each object type (HRU, landscape unit, channel, aquifer, reservoir and so on) and for the basin as a whole, by day, month, year and as an average over the simulation. Which files are written is set in `print.prt`: each row names a group of outputs and has a `y` or `n` for the daily, monthly, yearly and average annual files. A few files are written in every run.

The files go to the output folder named in `file.cio`, or to the run folder when none is named. Every run also writes `files_out.out`, a list of the output files it opened, which is the quickest way to see what a given `print.prt` produced.

Pages

Page	Content
Naming convention	How files are named and laid out, print timing, and the `print.prt` row behind every file.
Water balance	`hru_wb`, `hru-lte_wb`, `lsunit_wb` and `basin_wb`: precipitation, evapotranspiration, runoff, percolation, soil water and snow.
Channel, sediment, nutrient	`channel_sd`, `channel_sdmorph`, `sd_chanbud`, the sediment budget files `ch_sedbud`, `chanbud_order` and `sedbud`, their basin sums, and the legacy `channel` files.
Hydrographs	`hydout`, `hydin`, `deposition` and `ru`: water and loads passed between objects, and the `hydcon.out` connection list.
Recall	`recall` and `basin_psc`: loads released by point sources, inlets and diversions.
Nutrient balance and losses	`hru_nb`, `hru_ls`, `lsunit_nb`, `lsunit_ls`, `basin_nb`, `basin_ls` and `hru-lte_ls`: nitrogen and phosphorus balances and the sediment and nutrient losses from the landscape.
Aquifer, reservoir, plant	`aquifer`, `reservoir`, `wetland`, crop yields and the plant and weather (`_pw`) files.
Carbon	Soil, residue and plant carbon files.
gwflow	Outputs of the gridded groundwater module.
Checker and diagnostics	Files written in every run: `checker.out`, `diagnostics.out`, `files_out.out` and others.
Post-processing	The Python scripts in the `test` folder of the SWAT+ source.

Recent changes

- 62.0.1 adds the channel sediment budget files `ch_sedbud_*` (each channel), `chanbud_order_*` (each stream order) and `sedbud_*` (basin), all switched on by the `channel_sd` row.
- 62.0.0 added the carbon rows of `print.prt` (`hru_cb_gl`, `hru_cb_trf`, `lsu_cb_gl` and others) for the carbon output families; see Carbon.

Sources

- `print.prt` reader: `basin_print_codes_read.f90`
- Output folder: `output_path_module.f90`
- File opening: `proc_open.f90`, `output_landscape_init.f90`, `header_write.f90`, `header_sd_channel.f90`
- `files_out.out`: `proc_bsn.f90`